From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

FIITJEE

JEE(Advanced), 2014

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – V
PAPER-2

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	B	B	B
2.	B	B	A
3.	B	A	D
4.	D	D	B
5.	A	B	A
6.	A	D	A
7.	D	C	A
8.	A	D	A
9.	D	A	A
10.	B	B	C
11.	B	D	A
12.	C	A	C
1.	(A) → (q) (B) → (p, q, s, t) (C) → (p, q, s, t) (D) → (r, s)	(A) → (q, s) (B) → (r, t) (C) → (p, s) (D) → (q)	(A) → (p, q) (B) → (p, s) (C) → (p, r, t) (D) → (t)
2.	(A) → (s) (B) → (q, s) (C) → (p, r, t) (D) → (p, r, t)	(A) → (r, t) (B) → (s) (C) → (q) (D) → (p)	(A) → (r) (B) → (s) (C) → (q) (D) → (p)
1.	2	3	2
2.	2	4	1
3.	5	2	2
4.	3	3	5
5.	4	6	5

Physics

PART – I

SECTION – A

1. Let just after collision between A and B, v_1 and v_2 be the velocity of masses A and B respectively

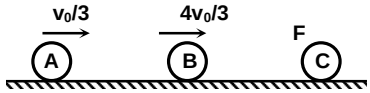
$v_0 = v_2 - v_1$ . . . (1)

$mv_0 = mv_1 + \frac{m}{2}v_2 \Rightarrow 2v_0 = 2v_1 + v_2$ . . . (2)

Using (1) and (2) we have

$v_1 = v_0/3$, $v_2 = 4v_0/3$

Now let v_3 and v_4 will be the velocity of mass B and respectively just after collision between B and C



Situation after collision between A and B

$4v_0/3 = [v_4 - v_3] \times M$ . . . (3)

$2mv_0/3 = mv_3/2 + Mv_4$ . . . (4)

Using $M \times (3) + (4)$

$$\frac{2v_0}{3}[m - 2M] = \frac{1}{2}(m + 2M)v_3$$

$$v_3 = \frac{4v_0}{3}\left[\frac{m - 2M}{m + 2m}\right]$$

For only one collision between masses A and B

$v_3 > 0$ and $v_3 \geq v_0 / 3$

$\Rightarrow m > 2M$ and $m \geq 10M/3$

2. $\lambda N(t) = \lambda\left(\frac{1}{2}\right)^n N_0$

Where n is no of half lives

5. $Q_A = 20\ \mu C$

$Q_B = 60\ \mu C$

$Q_C = 80\ \mu C$

12. $U_0 = nc_vT_A = 2 \times \frac{3}{2}RT_A$

$$\Rightarrow T_A = \frac{U_0}{3R},\ T_B = T_C = \frac{2U_0}{3R}$$

$$V_A = V_B = \frac{nRT_A}{P_A} = \frac{2U_0}{3P_0},\ V_c = \frac{nRT_C}{P_C} = \frac{4V_o}{3P_0}$$

$W_{cycle} = W_{AB} + W_{BC} + W_{CA}$

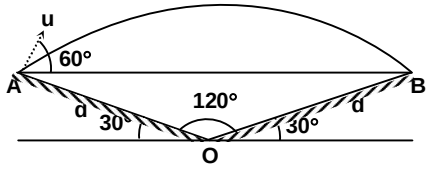
$= \frac{0.772U_0}{3}$

$$\%\eta = \frac{W_{cycle}}{\text{Heat supplied}} \times 100 = \frac{W_{cycle}}{Q_{BC} + Q_{AB}} \times 100$$

= 13.38%

SECTION - C

2. Since $\alpha = \beta$
AO = BO = 20 cm



$AB = \sqrt{d^2 + d^2 - 2d \cdot d\cos 120°}$

$AB = \sqrt{3}d$

Now AB is the range of the projectile from A to B.

$\sqrt{3}d = \dfrac{u^2 \sin 2\theta}{g}$

$\Rightarrow \sqrt{3}d = \dfrac{u^2\sqrt{3}}{2g}$

Putting d = 0.2 m

u = 2 m/s

4. $\rho u^{\eta} = C$

$\Rightarrow V^{-1}T^{\eta} = C$

$\Rightarrow TV^{-1/\eta} = C$

$\Rightarrow PV^{1-(1/\eta)} = C$

Compare with $PV^x = C$

$\Rightarrow x = 1 - \dfrac{1}{\eta}$

Now, $\Delta Q \propto \left(\dfrac{R}{r-1} + \dfrac{R}{1-x}\right)$ and $\Delta W \propto \left(\dfrac{R}{1-x}\right)$

$\Rightarrow \dfrac{\Delta W}{\Delta Q} = \dfrac{\left(\dfrac{R}{1-x}\right)}{\dfrac{R}{r-1} + \dfrac{R}{1-x}} = \left(\dfrac{r-1}{r-x}\right) = \dfrac{2}{3}$

$= \dfrac{\dfrac{5}{3} - 1}{\dfrac{5}{3} - x} = \dfrac{2}{3}$

$\Rightarrow x = 2/3 = 1 - \dfrac{1}{\eta}$

$\Rightarrow \eta = 3$

5. Power = 40 kV 10 mA = 400 W.

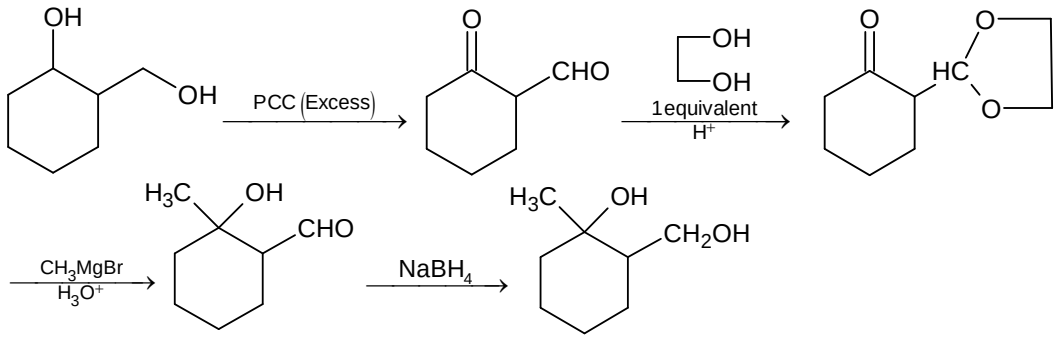
99 % of it is produced as heating power

Heating power = 396 W = 396 Joules per second.

Chemistry

PART – II

SECTION – A

1. 

2. Bayer-Villiger oxidation.

3. $2C + H_2(g) \longrightarrow C_2H_2(g)$

$\Delta H = 310 - \left[(2 \times 94) + 98\right]$

= 24 kcal

4. $NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$

$20 \times \frac{1}{10} = 2$ $\quad$ $16 \times \frac{1}{10} = 1.6$ $\quad$ 0 $\quad$ 0

0.4 $\quad$ 0 $\quad$ 1.6

The resulting solution is a basic buffer.

$pOH = pK_b + \log\frac{[Salt]}{[Base]}$

$= 4.74 + \log\frac{1.6}{0.4}$

= 4.74 + log4

= 4.74 + 0.6020

= 5.342

pH = 14 – 5. 342

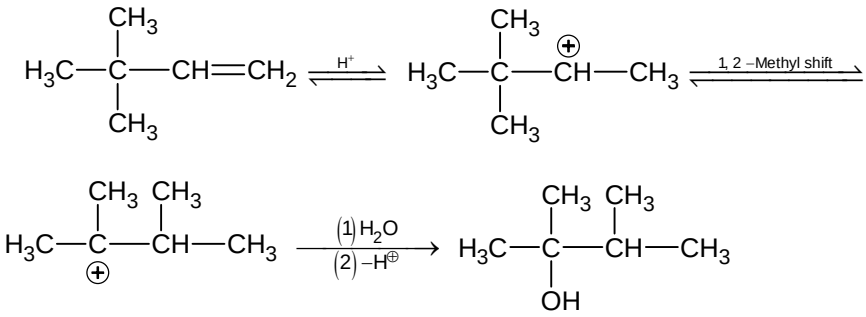
= 8.658

5. HF is a weaker acid than HI

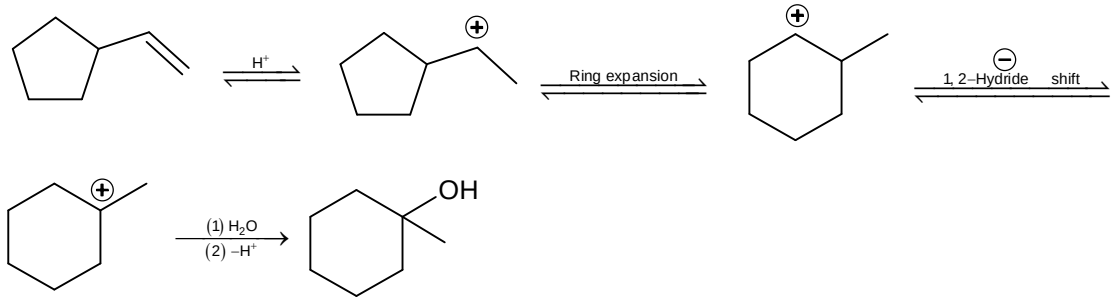
HF < HCl < HBr < HI

6. Li_2CO_3 is less stable than Rb_2CO_3.

7.



8.



9. Hydroboration-oxidation. Anti Markownikov's addition of H_2O.

10. $[Ag^+]$ to start the ppt. of Cl^-

$$= \frac{K_{sp}(AgCl)}{[Cl^-]} = \frac{1.7\times10^{-10}}{0.1} = 1.7\times10^{-9}\ M$$

$[Ag^+]$ to start the ppt. of Br^-

$$= \frac{K_{sp}(AgBr)}{[Br^-]} = \frac{5\times10^{-13}}{0.1} = 5\times10^{-12}\ M$$

$[Ag^+]$ to start the ppt. of CrO_4^{2-}

$$= \left[\frac{K_{sp}(Ag_2CrO_4)}{[CrO_4^{2-}]}\right]^{1/2} = \left[\frac{1.9\times10^{-12}}{0.1}\right]^{1/2} = 4.36\times10^{-6}\ M$$

Order of ppt. Br^-, Cl^-, CrO_4^{2-}.

11. When Cl^- starts precipitating

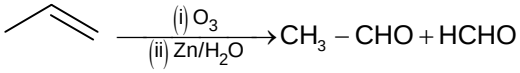
$$[Br^-] = \frac{K_{sp}(AgBr)}{[Ag^+]} = \frac{5\times10^{-13}}{1.7\times10^{-9}} = 3\times10^{-4}\ M$$

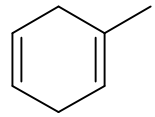
12. $[Ag^+]$ to start the ppt. of AgBr

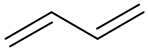
$$= \frac{K_{sp}(AgBr)}{[Br^-]} = \frac{5\times10^{-13}}{0.1} = 5\times10^{-12}\ M$$

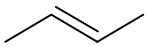
SECTION-B

1. (A)



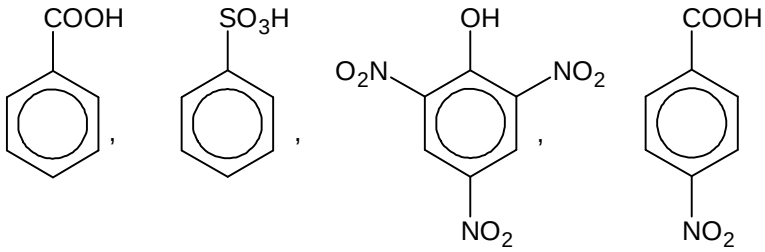
(B)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} CH_3-\overset{O}{\overset{\|}{C}}-CH_2-CHO+OHC-CH_2-CHO$

(C)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} HCHO+\underset{CHO}{\overset{CHO}{|}}+HCHO$

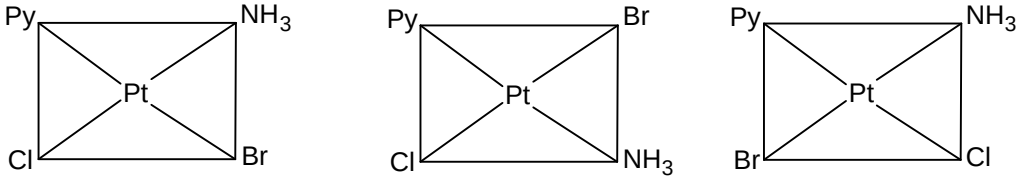
(D)  $\xrightarrow[(ii)\ Zn/H_2O]{(i)\ O_3} CH_3-CHO+CH_3-CHO$

SECTION – C

1. $H_2C(Cl)-CH_2-CH_2-CH_3$, $H_2C-\overset{Cl}{\overset{|}{C}}(H)-CH_2-CH_3$ $(d+\ell)$

2. 

3. $H_3C-\overset{O}{\overset{\|}{C}}-H$ and $H_3C-CH_2-\overset{O}{\overset{\|}{C}}-H$ react with Fehling solution.

4. 

5. $C_xH_y+\left(x+\frac{y}{4}\right)O_2 \longrightarrow xCO_2+\frac{y}{2}H_2O(\ell)$

5 ml	30 ml	0
0	$30-5\left[x+\frac{y}{4}\right]$	5x

$5x = 10,\ x = 2$

$5\left(x+\frac{y}{4}\right)=15$

$y = 4$

Formula of hydrocarbon = C_xH_y

$x + y = 2 + 4 = 6.$

Mathematics PART – III

SECTION – A

1. $\frac{dy}{dx} + y^3(1+\cos x) - \frac{y}{x} = 0$

$\frac{1}{y^3}\frac{dy}{dx} - \frac{1}{y^2 x} = -(1+\cos x)$

Let $\frac{-1}{y^2} = u$

$\therefore \frac{1}{2}\frac{du}{dx} + \frac{u}{x} = -(1+\cos x)$

$IF = e^{\int \frac{2}{x} dx} = x^2$

$\therefore ux^2 = -2\int x^2(1+\cos x)dx$

$\Rightarrow \frac{x^2}{2y^2} = \frac{x^3}{3} + x^2 \sin x + 2x\cos x - 2\sin x + k$

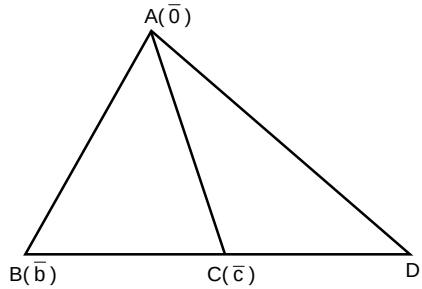
2. Put $x + \frac{1}{x} = t$

3. The p.v. of D = $3\vec{c} - 2\vec{b}$

AD $\perp$ AB $\Rightarrow \vec{b} \cdot (3\vec{c} - 2\vec{b}) = 0$

$\Rightarrow \cos A = \frac{2|\vec{b}|}{3|\vec{c}|} = \frac{1}{2}$

$\Rightarrow A = 60°$



4. Let number of men be 'm'.

Number of games played by men among themselves is $2 \times {}^{m}C_2 = m(m-1)$

Number of games which the men played with each of two women is 2m.

$\therefore m(m-1) - 2(2m) = 66$

$m^2 - 5m - 66 = 0$

$m = 11, -6$

$\therefore$ Total number of participants = 13 players

Number of games is $2 \times {}^{13}C_2 = 156$

5. Let $z = \alpha$ (real root)

$\alpha^3 + (3+i)\alpha^2 - 3\alpha - (m+i) = 0$

$\alpha^3 + 3\alpha^2 - 3\alpha - m = 0 \Rightarrow \alpha^2 - 1 = 0$

$\therefore$ m = 1, 5 $\Rightarrow \alpha = \pm 1$

6. $T_n = \sqrt{1+\frac{1}{n^2}+\frac{1}{(n+1)^2}} = \sqrt{\frac{n^4+2n^3+n^2+n^2+2n+1+n^2}{n^2(n+1)^2}} = \frac{n^2+n+1}{n(n+1)} = 1+\frac{1}{n(n+1)}$

7. $g'(x) = 4x^3 - 6x^2 + 2x$

$= 2x(2x^2 - 3x + 1)$

$= 2x(2x-1)(x-1)$

$\therefore a_1 = 0,\ a_2 = \frac{1}{2},\ a_3 = 1$

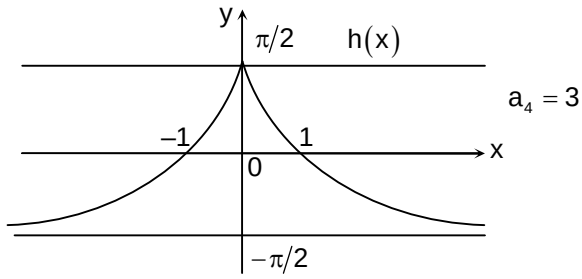
$g\left(\frac{1}{2}\right) = \frac{1}{16} - \frac{2}{8} + \frac{1}{4} + 4 = \frac{1}{16} + 4$

$\therefore BC = \frac{1}{16}$

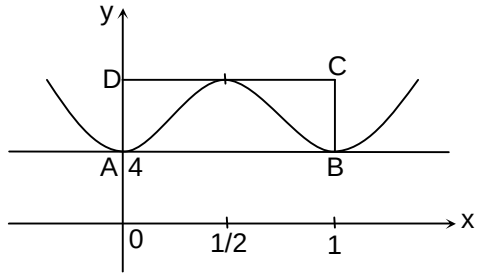
$\therefore$ Area of ABCD is = $\frac{1}{16} \times 1 = \frac{1}{16}$

Area = $2\int_0^1 \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right)dx$

Substitute $x = \tan\theta$ and integrate.



$g(x) =$



10. $\left(x+\frac{1}{x}\right)^2 + 2a\left(x+\frac{1}{x}\right) - 1 = 0$ ..... (1)

$\Rightarrow t^2 + 2at - 1 = 0$ ..... (2)

$t \geq 2$ and $t \leq -2$ (Given)

Equation (1) will have at least two positive roots, when at least one root of equation (2) will be greater than 2

11. Equation (1) will have at least two negative roots, when at least one roots of equation (2) will be less than –2

12. –2 and 2 must lie between roots

SECTION – B

1. $a^3 + b^3 + c^3 = p$

$a^3b^3 + b^3c^3 + c^3a^3 = q$

$a^3b^3c^3 = r$

(A) $\therefore \frac{a^3+b^3+c^3-3abc}{8R^3} = 8\Delta^3$

$\Rightarrow \frac{\left(p - 3r^{\frac{1}{3}}\right)64\Delta^3}{8(abc)^3} = 8\Delta^3$

$\Rightarrow p = r + 3r^{\frac{1}{3}}$

(B) $\dfrac{a^3+b^3+c^3}{4R^2}=14\Delta^2$

$\dfrac{\left(a^3+b^3+c^3\right)\left(16\Delta^2\right)}{4(abc)^2}=14\Delta^2$

$\dfrac{p}{r^{\frac{2}{3}}}=\dfrac{14}{4}$

$p=\dfrac{14}{4}r^{\frac{2}{3}}$

(C) $\dfrac{abc}{8R^3}=2\Delta^3$

$\Rightarrow \dfrac{abc\left(\Delta^3\right)64}{8(abc)^3}=2\Delta^3$

$\Rightarrow \dfrac{8}{r^{\frac{2}{3}}}=2$

$\Rightarrow r^{\frac{2}{3}}=4$

$\Rightarrow r=8$

(D) $a+b+c-2\left(a\sin^2A+b\sin^2B+c\sin^2C\right)=2\left(s-\Delta^2\right)$

$\left(a\sin^2A+b\sin^2B+c\sin^2C\right)=\dfrac{2\Delta^2}{2}$

$\Rightarrow \dfrac{p}{r^{\frac{2}{3}}}.4\Delta^2=\Delta^2$

$\Rightarrow p=\dfrac{r^{\frac{2}{3}}}{4}$

$\Rightarrow p=1,\ r=8$

2 **(A)** Let $(n-k)$ tail occur at any $n-k$ places

x T x T x T x T x T x ……. x T

Blank spaces between $(n-k)$ tails can be occupied by k heads.

Total probability = ${}^{n-k+1}C_k\left(\dfrac{1}{2}\right)^k\left(\dfrac{1}{2}\right)^{n-k}=\dfrac{{}^{n-k+1}C_k}{2^n}$

(B) Treating the k heads as one unit,

Probability = $\dfrac{{}^{n-k+1}C_1}{2^n}$

(C) and **(D)** follows from **(A)** and **(B)**

SECTION – C

1. Let OD = $\sqrt{a}$ and AD = $\sqrt{1-a}$

OC = OD + CD = $\sqrt{1-a}$

$OC^2 = \left(\sqrt{a} + \sqrt{1-a}\right)^2 = 1 + 2\sqrt{a(1-a}$

$a(1-a)$ should be maximum

$\Rightarrow$ at $a = \frac{1}{2}$, the above expression attains maximum value

$\therefore$ OD = $\frac{1}{\sqrt{2}}$

$\therefore$ AD = $\frac{1}{\sqrt{2}}$

AB = 2AD = $\sqrt{2}$

2. $\cos B = \frac{1}{2} = \frac{a^2 + c^2 - b^2}{2ac}$

$\Rightarrow (a-c)^2 - b^2 + ac = 0$

$\Rightarrow$ 1 satisfies the given equation

3. The portion of tangent intercepted between tangent at vertices subtends right angle at focus.
$\therefore$ Centre of circle is the midpoint of the portion of the tangents intercepted between the tangent at vertices

4. Let the six persons P_1, P_2, P_3, P_4, P_5 and P_6 be seated.

$\underset{x_1}{_} P_1 \underset{x_2}{_} P_2 \underset{x_3}{_} P_3 \underset{x_4}{_} P_4 \underset{x_5}{_} P_5 \underset{x_6}{_} P_6 \underset{x_7}{_}$

x_1 is the number of seats to left of P_1, x_2 is the number of seats in between P_1 and P_2, ……, x_7 is the number of seats to right of P_6 such that $x_1 + x_2 + x_3 + ... + x_7 = 9$.

i.e., coefficient of x^9 in $\left(1 + x + x^2 + ...\right)^2 \left(x + x^2 + x^3 + ...\right)^5 = {}^{15-6+1}C_6 = {}^{10}C_6$

$\because$ 6 persons can be arranged in 6! ways

$\therefore$ ${}^{10}P_6$ is the answer

5. $6^{83} + 8^{83} = (7+1)^{83} + (7-1)^{83}$
$= 2({}^{83}C_0 (7)^{83} + {}^{83}C_2 (7)^{81} + \ldots + {}^{83}C_{82} (7)^1)$
$= 7\lambda + 83 \times 7 \times 2$
$\Rightarrow$ Remainder is 5